Ethan McKeen

North Vancouver, BC

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EDUCATION

University of British Columbia — **Completed: April, 2025**

Masters of Engineering **–** *Electrical and Computer Engineering Department* — Grade: 4.0 GPA

- Focused on Machine Learning and Software Engineering

University of British Columbia — **Completed: April, 2024**

Bachelors of Applied Science **–** *Electrical and Computer Engineering Department* — Grade: 3.7 GPA

WORK EXPERIENCE

Quadrant Electronics Inc, North Vancouver, BC — **Apr, 2025 – Present**

Software, Firmware and ML Engineer

- Made **UI and SQL Azure database improvements** to a custom inventory control software.
- Wrote **firmware in C++ and MicroPython** for an ESP32 microcontroller including functionality for I2C, SPI, wifi communications, and multithreading; conducted benchmark testing to validate and improve high-speed network communication.
- Modeled RF power amplifier impairments using various techniques including the development of a non-linear **retentive neural network** and a **memoryless digital signal processing** approach.

NeuroPrior AI, North Vancouver, BC — **Apr, 2024 – Sep, 2024**

Software and Firmware Engineer

- Collaborated in a 5-person team to design a medical EEG/ECG wearable device.
- Developed **firmware in C for BLE communication** (advertising, connecting, ADC data transmission).
- Applied **DSP** to transform sensor input into real-time audio playback and FFT visualizations in both Python and MATLAB.
- Built a Python **desktop application** with a **GUI** to automate device connectivity, stream signals, and visualize data.

Quadrant Electronics Inc, North Vancouver, BC — **Apr, 2022 – Aug, 2022**

Junior Software Engineer

- Designed and prototyped a **VR pilot training program** in Unity with interactive cockpit controls and instrument simulations for fault-scenario training.
- Created an **AR mobile application** to extend training accessibility.
- Gained experience in **3D data** interaction, rendering, and visualization.
- Engineered workflows **independently** from concept to working prototype, showcasing initiative and problem solving skills.

Celly Technologies, North Vancouver, BC — **Apr, 2020 – Aug, 2020**

Junior Software and Firmware Developer

- Worked in a 4-person team to design a mobile application that monitored sump pump status and provided real-time user alerts.
- Wrote an **Android app** using OpenCV for **image/video frame processing**, automating LED detection and event classification.
- Integrated AWS backend services **to manage user data** and make it accessible online, ensuring reliable and scalable data delivery.

RESEARCH

Investigating the Role of Test Cases in LLM-Based Python and JavaScript Code Translation

- Investigated different **prompt engineering** and **feature engineering** approaches and **iterative post processing techniques** for two-way Python and JavaScript **code translation**. Our improvements increased the LLM translation success rate from **44% to 91%.**
- Tests included **generating synthetic test cases** to translate code with different LLM models including **Gemini-2.0** and **GPT-3.5**.

Vision Transformer Based Chessboard Image Analysis

- Explored the use of **Vision Transformers** for classifying digital chess board pieces for game state extraction.
- Involved extensive **data pre-processing** and was able to significantly **reduce training complexity** and time while maintaining a model **generalizable** to many board styles and colors. For **100% classification accuracy** the training data required was reduced to **500 from 80,000.**

Transformer Based Actor Critic Agent

- Investigated use cases of **transformers in a reinforcement learning** setting. Tests were done using an actor critic model.
- Utilized **OpenAI Gym** to test the model's performance in different environments including Cart Pole and lunar Lander.
- Transformer model took longer to converge but was more stable. Reduced agent failure rate from **0.02% to 0.0%.**

Languages		Tools			Software	
• Python	• Java	• LLaMA	• PyTorch	• OpenCV	• Unity	• SolidWorks
• C and C++	• JavaScript	• Git	• TensorFlow	• Sckit-Learn	• Matlab	• Android Studio